

Software Requirements Specification

Inventory Management System (IMS)

Version 1.0

Prepared in accordance with IEEE Std 830-1998

Classification: Confidential — Internal Use Only

1. Introduction

This document defines the comprehensive software requirements for the Inventory Management System (IMS). This Software Requirements Specification (SRS) serves as an authoritative reference for all stakeholders during the design, development, implementation, and testing phases of the system. The final product will contain only the features and functionalities described herein. Any additional functionality will require a formal change request, following which a revised release of this document and the product will be produced.

1.1 Purpose

This specification document describes the capabilities that will be provided by the Inventory Management System (IMS). It also states the various constraints by which the system will abide. The IMS is designed to replace or supplement traditional manual inventory management practices by offering a simple, efficient, and easy-to-use solution. With every order processed, the system automatically generates an invoice and updates stock levels, making the system faster and reducing dependence on manual intervention. The integration of Power BI further enables detailed sales analysis and inventory insights for decision-makers.

The intended audience for this document includes the development team, testing team, project managers, and end users of the product.

1.2 Document Conventions

This SRS follows the IEEE 830-1998 standard for software requirements documentation. All functional requirements are uniquely identified using a standardised tag format (e.g., REQ-1, REQ-2) to ensure traceability throughout the project lifecycle.

User interface elements are represented in bold text, while technical terms are italicised and explained in the Definitions section. Feature priorities are categorised as High, Medium, or Low, where High-priority features are critical for core system functionality and must be implemented in the initial release.

1.3 Intended Audience and Reading Suggestions

This document is intended for the following audience:

- **Developers:** Should focus primarily on Sections 3 (External Interface Requirements) and 4 (System Features) to understand system behaviour and implementation logic.
- **Project Managers:** Should review Section 2 (Overall Description) to ensure alignment with project objectives, scope, and delivery timelines.
- **Testers and Evaluators:** Should refer to Section 5 (Non-Functional Requirements) to define testing criteria related to performance, security, and reliability.
- **Stakeholders and Reviewers:** May focus on Section 2.2 (Product Functions) to verify that business and organisational needs are adequately addressed.

Readers are encouraged to begin with the Introduction sections before proceeding to the detailed functional requirements.

1.4 Product Scope

The Inventory Management System (IMS) is a Windows-based desktop application designed to manage the complete inventory operations of a retail store. The system enables users to process incoming orders, automatically generate GST-compliant invoices in PDF format, and update stock levels in real time — significantly reducing manual workload on store employees.

Administrative and managerial users are provided with Power BI dashboards that offer insights into sales trends, product demand, inventory levels, and customer demographics. The system is designed to be efficient, reliable, and scalable for retail operations handling multiple concurrent users.

1.5 References

- IEEE Std 830-1998 — Recommended Practice for Software Requirements Specifications
- User Interface Style Guide for Organisational Desktop Applications
- Power BI Documentation — Microsoft
- Node.js Official Documentation — <https://nodejs.org>
- PostgreSQL Official Documentation — <https://www.postgresql.org>

1.6 Definitions, Acronyms, and Abbreviations

The following abbreviations and definitions have been used throughout this document:

Term / Acronym	Definition
IMS	Inventory Management System
SRS	Software Requirements Specification
IEEE	Institute of Electrical and Electronics Engineers
GST	Goods and Services Tax
PDF	Portable Document Format
Power BI	Business Intelligence and data analytics tool by Microsoft
PostgreSQL	Open-source relational database management system used as the backend database
Node.js	JavaScript runtime environment used for backend processing
React.js	JavaScript library used for building the desktop application frontend interface
Admin	Administrator — highest privilege user role in the system
HOD	Head of Department (General Manager in the context of IMS)
SKU	Stock Keeping Unit — unique identifier for each product
RDBMS	Relational Database Management System
REQ	Requirement — used as a prefix for requirement identifiers (e.g., REQ-1)

1.7 Overview

The remainder of this SRS document describes the overall system description, external interface requirements, system features, non-functional requirements, and use case descriptions in detail. Each section builds upon the previous to provide a complete picture of the system from both functional and non-functional perspectives.

2. Overall Description

The IMS is developed specifically to manage the inventory of a retail store, reducing the workload of manual stock-keeping and billing operations. The system automatically processes incoming orders, generates GST-compliant invoices, updates stock data, and provides analytical insights through Power BI dashboards.

2.1 Product Perspective

The Inventory Management System is a Windows-based, self-contained, and independent desktop application. It does not depend on any external third-party application for its core operations. The system connects to a PostgreSQL database hosted on a Windows server for persistent data storage. Power BI is integrated as the analytics and reporting layer. The system is intended for internal use within a retail store environment and is not exposed as a public-facing application.

2.1.1 System Interfaces

The IMS interfaces with the following external systems:

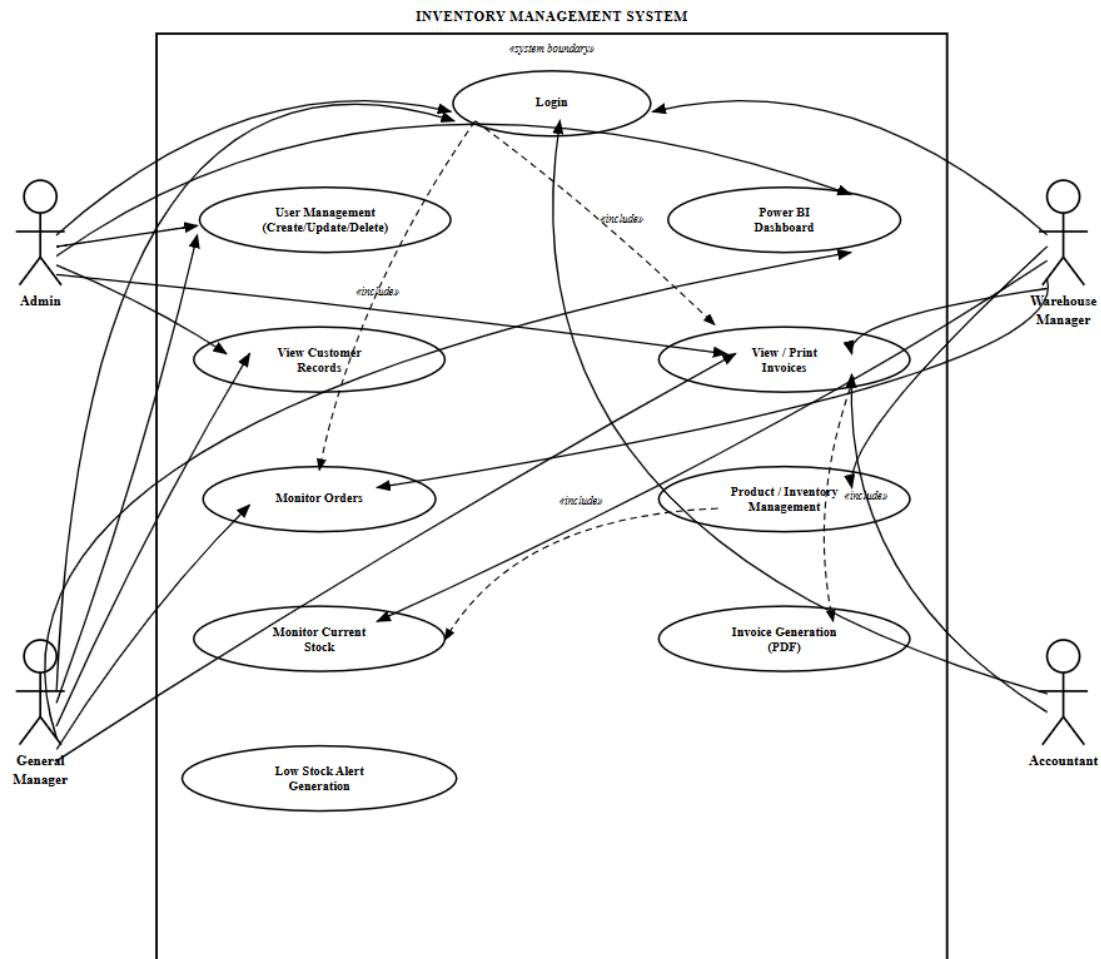
- PostgreSQL Database Server: All persistent data including product information, customer records, order data, invoice records, and user accounts are stored in and retrieved from the PostgreSQL database via standard database connection drivers.
- Power BI: The system feeds structured sales, inventory, and customer data to Power BI for analytical dashboards and reporting purposes.
- Python PDF Generation Module: A Python-based backend module is invoked for generating printable, GST-compliant invoice PDFs from order data.

2.1.2 User Interfaces

The application will have a Graphical User Interface (GUI) with a menu-driven, role-based screen layout built using React.js rendered in an Electron desktop shell. The following screens will be provided:

1. A Login Screen for entering the username and password. The system will determine the role of the user upon login and display only the screens and functions relevant to that role.
2. A User Management Screen (accessible to Admin and General Manager) for creating, modifying, and deleting user accounts and assigning roles.
3. A Product / Inventory Management Screen (accessible to Warehouse Manager) for adding, updating, and deleting product records and monitoring current stock levels.
4. An Invoice Management Screen (accessible to Accountant, Warehouse Manager, General Manager, and Admin) for viewing, generating, and printing GST-compliant invoices in PDF format.
5. A Low Stock Alerts Screen (accessible to Warehouse Manager) displaying products that have fallen below the defined reorder threshold.

6. A Power BI Dashboard Screen (accessible to Admin and General Manager) for viewing sales trends, inventory analytics, product demand, and customer demographic insights.
7. A Customer Records Screen for viewing stored customer data (name, contact, address, age, purchase history) used for Power BI analytics.



2.1.3 Hardware Interfaces

The following minimum hardware specifications are required for the proper functioning of the system:

- Screen resolution of at least 1024×768 pixels is required for proper viewing of all application screens. Higher resolutions are fully supported.
- A standard keyboard and mouse for user input.
- A printer (inkjet, laser, or dot matrix) with appropriate drivers installed and connected to the system for printing of invoices and reports.
- A minimum of 8 GB RAM on the server machine hosting the PostgreSQL database. The client machines should have a minimum of 4 GB RAM.
- A minimum of 50 GB of available storage on the server for the database and application files.

2.1.4 Software Interfaces

- Operating System: Windows 10 or above on both client and server machines.
- Database: PostgreSQL 14 or above as the Relational Database Management System (RDBMS). The system must provide database table definitions for setup by the database administrator.
- Backend Runtime: Node.js (LTS version) for core backend processing and API handling.
- PDF Generation: Python 3.x with ReportLab or equivalent library for generating GST-compliant invoice PDFs.
- Frontend Framework: React.js bundled within an Electron.js shell for the desktop GUI.
- Analytics: Power BI Desktop / Power BI Service for dashboards and reporting.

Note: Backend development tools (Node.js, Python, React.js, Electron) are required during development. The final application will be packaged as a standalone Windows installer (.exe setup) to be delivered to the client.

2.1.5 Communications Interfaces

None. The system operates as a standalone Windows desktop application communicating with a local or LAN-hosted PostgreSQL server. No internet-based communication interfaces are required for core functionality. Power BI may require internet connectivity for cloud-based features, but this is external to the IMS itself.

2.1.6 Memory Constraints

The server machine hosting the PostgreSQL database should have a minimum of 8 GB RAM and at least 50 GB of available disk space on an SSD or equivalent storage medium. Client machines running the IMS desktop application should have a minimum of 4 GB RAM and 2 GB of available disk space for the application installation.

2.1.7 Operations

This product release will not cover any automated database housekeeping aspects. The database administrator at the client site will be responsible for refreshing, purging, and archiving old data in accordance with the organisation's data retention policy. Database backup and recovery will also be handled by the DBA.

However, the system will provide an administrative 'Reset System' function (accessible to Admin only) that will, upon explicit confirmation, delete all transactional data from the database while preserving user accounts and product master records.

2.1.8 Site Adaptation Requirements

8. New database table definitions created for this system must be installed on the organisation's PostgreSQL server prior to first use.
9. Product information, customer records, and user account information must be entered into the system before order processing and invoice generation can begin.
10. Power BI must be configured and connected to the PostgreSQL data source before dashboards can be made operational.
11. All client machines must meet the minimum hardware and software specifications outlined in Sections 2.1.3 and 2.1.4.
12. A stable local area network (LAN) connection must be available between client machines and the database server.

2.2 Product Functions

The system will allow access only to authorised users with specific roles (Admin, General Manager, Warehouse Manager, Accountant). Depending on the user's role, he/she will be able to access only the specific modules of the system assigned to that role.

A summary of the major functions the software will perform:

13. A login facility enabling only authorised access into the system.
14. Admin and General Manager will be able to create, modify, and delete user accounts with specific roles.
15. Admin will be able to reset the system, leading to deletion of all transactional data from the backend database.
16. Warehouse Manager will be able to add, modify, and delete information about products in inventory, including stock quantities and pricing.
17. The system will automatically update stock levels upon processing of each order.
18. The system will automatically generate a GST-compliant invoice in PDF format upon receipt of order data (customer details and items ordered).
19. Admin, General Manager, Warehouse Manager, and Accountant will be able to view and print invoices.
20. The system will automatically trigger low stock alerts to the Warehouse Manager when any product's stock level falls below the defined reorder threshold.
21. Admin and General Manager will have access to Power BI dashboards displaying sales data, inventory levels, product demand, and customer demographics.
22. Customer data (name, contact, address, age, purchase history) will be stored in the backend for Power BI analytics and order-processing purposes.

2.3 User Classes and Characteristics

The following user roles are defined for the IMS. Each role has a distinct set of permissions and responsibilities within the system:

User Role	Responsibilities	System Access	Technical Expertise Required
Admin	Full system control. Manages all user accounts, resets the system, and has access to all modules.	All modules including User Management, Inventory, Invoices, Power BI, and System Reset.	Should have working knowledge of Windows OS, PostgreSQL, and system administration.
General Manager	Oversees operations. Views Power BI dashboards, manages users, and monitors sales and inventory data.	User Management, Invoice viewing, Power BI Dashboards, Customer Records.	Should be comfortable using desktop applications. No technical expertise required.
Warehouse Manager	Manages product inventory. Adds, updates, and deletes products. Monitors stock levels and responds to low-stock alerts.	Inventory/Product Management, Invoice viewing, Low Stock Alerts.	Should be comfortable using general-purpose desktop applications. Familiarity with retail inventory preferred.
Accountant	Handles billing and financial records. Views invoices and order data. Cannot modify stock or user accounts.	Invoice viewing and printing only. Read-only access to order records.	Should be comfortable using desktop applications. Basic computer literacy required.
Customer (External)	External entity. Not a system user. Customer data (name, contact, address, age, purchase history) is stored in the backend for invoice generation and Power BI analytics.	No system login. Data is captured and stored at the time of order placement.	Not applicable.

2.4 Operating Environment

The IMS shall operate as a Windows-based desktop application accessible on machines running Windows 10 or above. The backend database (PostgreSQL) will be hosted on a Windows server within the organisation's local area network. The system must support standard desktop environments and allow multiple users to access it concurrently over the LAN. The system must function reliably on both the server and client machines meeting the minimum hardware specifications outlined in Section 2.1.3.

2.5 Design and Implementation Constraints

- The software is intended to be installed and operated on Microsoft Windows 10 and above only.
- The RDBMS to be used is PostgreSQL 14 or above only.
- The system must comply with applicable data protection practices, ensuring that customer details, product data, and financial records are securely handled and not exposed to unauthorised users.
- Wherever possible, development shall utilise open-source technologies (Node.js, PostgreSQL, Python, React.js, Electron.js) to reduce licensing costs and improve flexibility.
- The application must be designed with scalability and maintainability in mind, allowing it to accommodate increased user load and future feature additions without major architectural changes.
- The Administrator will be responsible for implementing a security policy to safeguard inventory and billing data from unauthorised access or modification.

2.6 User Documentation

To assist users in effectively operating the system, the following documentation will be provided:

- An interactive FAQ document addressing common user queries and troubleshooting steps.
- Detailed user documentation explaining the complete workflows — from order receipt to invoice generation and stock update.
- Short video tutorials demonstrating key system operations, including login, inventory management, invoice generation, and Power BI dashboard navigation.

2.7 Assumptions and Dependencies

- It is assumed that a stable local area network (LAN) connection is available between client machines and the PostgreSQL server at all times during operation.
- It is assumed that the Power BI environment is set up and configured to connect to the PostgreSQL database before dashboards are made operational.

- It is assumed that customer data (name, contact, address, age) is available and pre-loaded into the system or captured at the point of order placement before invoice generation is initiated.
- User roles are predefined and assigned by the Admin or General Manager prior to first use.
- Any failure or disruption in the PostgreSQL server or LAN connectivity may impact system functionality and data availability.
- The number of simultaneous users is expected to be 4 to 5 at any given time in the current deployment phase. Support for a higher number of concurrent users is planned for a future release.

2.8 Apportioning of Requirements

The following requirements are identified as planned for future releases and are not part of the current version (Version 1.0) of the IMS:

- Support for a higher number of simultaneous users (beyond 4–5 concurrent users).
- Customer-facing login portal for order placement directly through the system.
- Supplier management module for tracking and managing supplier information and purchase orders.
- Integration with external e-commerce or point-of-sale platforms.

3. Specific Requirements

This section contains all the software requirements at a level of detail sufficient to enable designers to design a system to satisfy those requirements, and testers to test that the system satisfies those requirements.

3.1 External Interface Requirements

The following screens will be provided by the Inventory Management System:

Login Screen

This is the first screen displayed when a user wishes to access the system. It will allow the user access to different modules based on the user's role. Various fields available on this screen will be:

- 23. Username: Alphanumeric, minimum 3 characters, maximum 20 characters.
- 24. Password: Alphanumeric, minimum 8 characters, maximum 20 characters.

If the user enters a correct username and password, the system will display the relevant function menu based on the user's role.

Role-Based Function Menu Screen

Upon successful login, the system will display a menu relevant to the user's role:

- 25. Admin: User Management, Inventory Management (view), Invoice Management, Power BI Dashboard, System Reset.
- 26. General Manager: User Management, Invoice Management (view/print), Power BI Dashboard, Customer Records.
- 27. Warehouse Manager: Inventory/Product Management, Invoice Management (view/print), Low Stock Alerts.
- 28. Accountant: Invoice Management (view/print only). No access to inventory or user management.

Product / Inventory Management Screen

This screen is accessible only to the Warehouse Manager. It allows the user to add, modify, and delete product records and monitor current stock levels. Various fields available on this screen will be:

- 29. Product Code (SKU): Alphanumeric, unique for each product, system-generated or manually assigned, maximum 20 characters. Cannot be NULL.
- 30. Product Name: Alphanumeric string, maximum 100 characters. Cannot be NULL.
- 31. Product Category: Alphanumeric string, maximum 50 characters. Optional.
- 32. Unit Price (MRP): Decimal value, greater than zero. Cannot be NULL.

- 33. GST Rate (%): Decimal value representing the applicable GST percentage for the product. Cannot be NULL.
- 34. Current Stock Quantity: Integer, greater than or equal to zero. Cannot be NULL.
- 35. Reorder Threshold: Integer representing the minimum stock level below which a low stock alert is triggered. Cannot be NULL.

Invoice Management Screen

This screen is accessible to Admin, General Manager, Warehouse Manager, and Accountant for viewing and printing invoices. Invoice generation is triggered automatically by the system upon receipt of a valid order. Fields displayed on the invoice include:

- 36. Invoice Number: Unique, system-generated identifier. Cannot be NULL.
- 37. Invoice Date and Time: Auto-populated by the system at time of generation.
- 38. Customer Name, Contact Number, and Address: Retrieved from customer records.
- 39. Order Items: List of products ordered, with quantity and unit price.
- 40. GST Amount and Total Amount Payable: Automatically computed by the system based on applicable GST rates.

Low Stock Alerts Screen

This screen is accessible only to the Warehouse Manager. It displays a list of all products whose current stock level has fallen below the defined reorder threshold. For each such product, the screen displays the Product Code, Product Name, Current Stock Quantity, and Reorder Threshold.

Customer Records Screen

This screen is accessible to Admin and General Manager for viewing customer information. Customer data is captured at the time of order placement and stored in the backend. Fields maintained for each customer include:

- 41. Customer ID: Unique, system-generated identifier.
- 42. Customer Name: Alphabetic characters only, maximum 60 characters. Cannot be NULL.
- 43. Contact Number: Numeric, 10 digits.
- 44. Address: Alphanumeric, maximum 255 characters.
- 45. Age: Integer, between 1 and 120. Used for Power BI demographic analytics.
- 46. Purchase History: Linked order and invoice records associated with this customer.

Power BI Dashboard Screen

This screen is accessible to Admin and General Manager. It embeds the Power BI dashboard within the application interface, providing analytical views of:

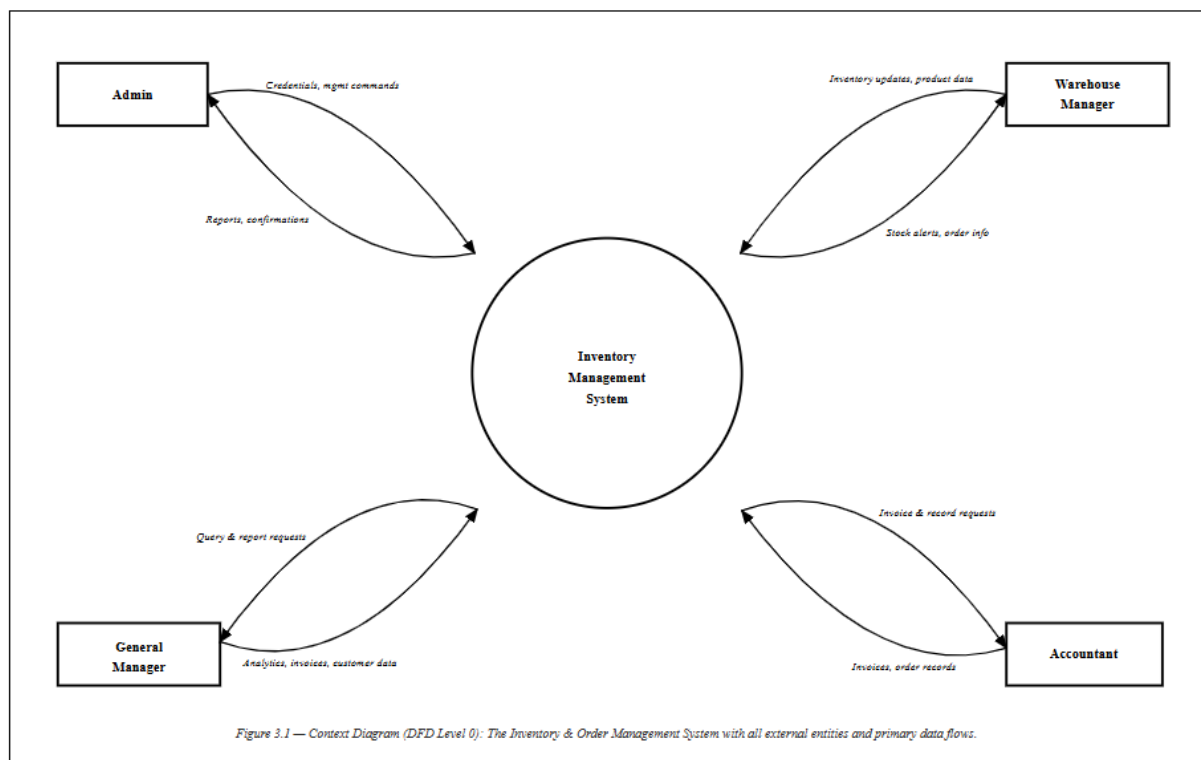
47. Sales trends over time (daily, weekly, monthly).
48. Product-wise demand and revenue analysis.
49. Current inventory status and stock movement.
50. Customer demographic insights (age groups, geographic distribution).
51. Profit margins per product.

User Management Screen

This screen is accessible to Admin and General Manager. It allows the creation, modification, and deletion of user accounts. Fields available on this screen include:

52. Full Name: Alphabetic characters, maximum 60 characters. Cannot be NULL.
53. Username: Alphanumeric, unique for every user account, maximum 20 characters. Cannot be NULL.
54. Password: Alphanumeric, minimum 8 characters, maximum 20 characters. Cannot be NULL.
55. Role: One of the following — Admin, General Manager, Warehouse Manager, Accountant. Cannot be NULL.

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3.2 System Features

3.2.1 User Authentication and Login

Description

The system will provide a secure login mechanism to ensure only authorised users can access the system. Upon successful login, the system will determine the user's role and display only the screens and functions appropriate to that role.

Validity Checks

- 56. Only users with a valid username and password registered in the system will be permitted to log in.
- 57. Username cannot be NULL and must be between 3 and 20 alphanumeric characters.
- 58. Password cannot be NULL and must be between 8 and 20 characters.
- 59. The system will display an appropriate error message for invalid credentials without revealing which field is incorrect, for security purposes.

Sequencing Information

A user account must be created by the Admin or General Manager before any user can log into the system. At initial system setup, only a default Admin account with the password 'Change_On_Install' will be present.

Error Handling / Response to Abnormal Situations

If the username or password entered is incorrect, the system will display an error message prompting the user to re-enter credentials. After three consecutive failed login attempts for the same account, the system will lock the account and notify the Admin.

3.2.2 User Account Management

Description

The system will maintain information about all users who are authorised to access the system. The following information will be maintained for each user: Full Name, Username, Password, Role.

The system will allow Admin and General Manager to add, modify, and delete user accounts, and to list all existing user accounts.

Validity Checks

- 60. Only users with the role Admin or General Manager will be permitted to access the User Account Management module.
- 61. Full Name cannot be NULL.
- 62. Username cannot be NULL. Must be unique for every user. Alphanumeric, maximum 20 characters.
- 63. Password cannot be NULL. Must be a minimum of 8 characters.
- 64. Role cannot be NULL. Must be one of: Admin, General Manager, Warehouse Manager, Accountant.

Sequencing Information

At system startup, only a default Admin account is present. The Admin must create user accounts for all other users before those users can access the system.

Error Handling / Response to Abnormal Situations

If any of the above validations fail, appropriate error messages will be displayed to the user. Duplicate usernames will not be permitted and the system will prompt the user to choose a different username.

3.2.3 Product / Inventory Management

Description

The system will maintain information about all products stocked by the retail store. The following information will be maintained for each product: Product Code (SKU), Product Name, Product Category, Unit Price (MRP), GST Rate (%), Current Stock Quantity, Reorder Threshold.

The system will allow the Warehouse Manager to add, modify, and delete product records, and to view the current inventory list.

Validity Checks

65. Only the user with the role Warehouse Manager will be permitted to access the Product Management module.
66. Product Code (SKU) cannot be NULL. Must be unique for each product. Maximum 20 alphanumeric characters.
67. Product Name cannot be NULL. Alphanumeric, maximum 100 characters.
68. Unit Price (MRP) cannot be NULL. Must be a positive decimal value greater than zero.
69. GST Rate (%) cannot be NULL. Must be a valid non-negative decimal value.
70. Current Stock Quantity cannot be NULL. Must be an integer greater than or equal to zero.
71. Reorder Threshold cannot be NULL. Must be a positive integer.

Sequencing Information

Product information must be entered into the system before order processing and invoice generation can be initiated.

Error Handling / Response to Abnormal Situations

If any of the above validations fail, appropriate error messages will be displayed to the user. Duplicate Product Codes will not be permitted.

3.2.4 Order Processing and Automatic Stock Update

Description

The system will receive order data from the backend — comprising customer details (name, contact, address, age) and a list of items ordered with their quantities. Upon receipt of this data,

the system will validate the order, process it, automatically update the stock levels for each ordered product, and trigger invoice generation.

Validity Checks

- 72. Each item in the order must correspond to a valid Product Code (SKU) existing in the product database.
- 73. The quantity ordered for each item must not exceed the available stock quantity for that product.
- 74. Customer name cannot be NULL.
- 75. Stock levels will be decremented automatically upon successful order processing. This action cannot be reversed manually.

Sequencing Information

Product information and customer records must be present in the system before an order can be processed.

Error Handling / Response to Abnormal Situations

If a product in the order does not exist in the system, the system will display an error message identifying the invalid product. If the requested quantity for any product exceeds available stock, the system will notify the relevant user and will not process the order until the issue is resolved.

3.2.5 Invoice Generation

Description

The system will automatically generate a GST-compliant invoice in PDF format upon successful processing of each order. The invoice will be generated by the Python-based PDF module invoked by the backend. The invoice will include all order details, applicable GST calculations, and the final amount payable.

Validity Checks

- 76. An invoice can only be generated after a valid order has been successfully processed.
- 77. GST must be calculated correctly based on the GST Rate (%) defined for each product.
- 78. Invoice Number must be unique and system-generated. Cannot be NULL.
- 79. All financial calculations (subtotal, GST amount, total payable) must be accurate to two decimal places.

Sequencing Information

Order processing must be completed before invoice generation is triggered. Product records with valid GST rates must be present in the system.

Error Handling / Response to Abnormal Situations

If the PDF generation module fails, the system will log the error and notify the user. The order record will be preserved in the database to allow for manual invoice re-generation.

3.2.6 Low Stock Alert Management

Description

The system will monitor stock levels for all products in real time. When any product's current stock quantity falls below its defined Reorder Threshold (as a result of order processing), the system will automatically generate a low stock alert for the Warehouse Manager.

Validity Checks

- 80. A low stock alert will be triggered only when the current stock quantity of a product is strictly less than its Reorder Threshold.
- 81. Alerts will be displayed only to the user with the role Warehouse Manager.
- 82. Each product will generate at most one active alert at a time. Duplicate alerts for the same product will not be generated until the stock is replenished above the threshold.

Sequencing Information

Product information including the Reorder Threshold must be entered into the system before low stock alerts can be generated.

Error Handling / Response to Abnormal Situations

If any of the above conditions are not met, the alert will not be generated. Appropriate messages will be displayed to the user where applicable.

3.2.7 Power BI Analytics and Reporting

Description

The system will feed structured data from the PostgreSQL database to Power BI for analytical reporting. Power BI dashboards will provide Admin and General Manager with visual insights into sales performance, inventory levels, product demand, and customer demographics. The Warehouse Manager will have access to stock-related alerts and inventory summary views.

Validity Checks

- 83. Only Admin and General Manager will have access to the full Power BI dashboards.
- 84. The data fed to Power BI must be current and reflect the latest order, stock, and customer data from the PostgreSQL database.

Sequencing Information

Order data, customer data, and inventory data must be present in the system before meaningful Power BI dashboards can be generated. Power BI must be configured and connected to the PostgreSQL data source prior to use.

Error Handling / Response to Abnormal Situations

If the Power BI connection is unavailable, the system will display an appropriate error message. Core IMS functions (order processing, invoice generation, inventory management) will continue to operate independently of the Power BI integration.

3.2.8 Customer Data Management

Description

The system will store and maintain customer information in the backend database for the purposes of invoice generation and Power BI analytics. The following information will be maintained for each customer: Customer ID, Customer Name, Contact Number, Address, Age, Purchase History (linked order and invoice records).

Validity Checks

- 85. Customer ID cannot be NULL. Must be unique for each customer. System-generated.
- 86. Customer Name cannot be NULL. Alphabetic characters only, maximum 60 characters.
- 87. Contact Number: Optional. When present, must be a numeric string of 10 digits.
- 88. Address cannot be NULL. Alphanumeric, maximum 255 characters.
- 89. Age: Optional. When present, must be an integer between 1 and 120.

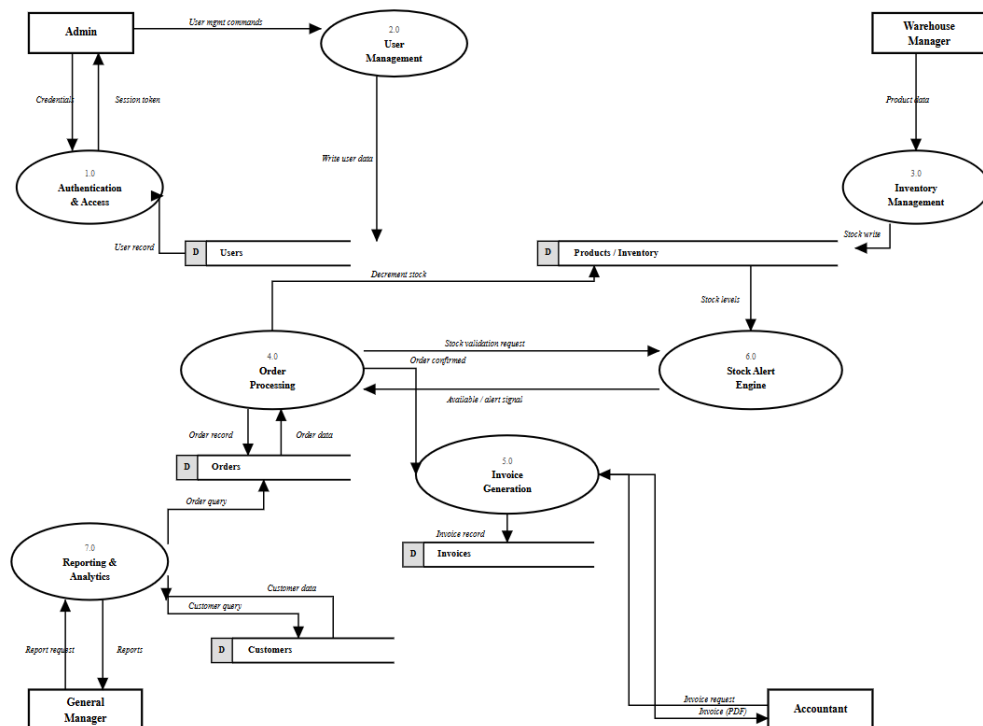
Sequencing Information

Customer data must be present in the system before order processing and invoice generation can be initiated for that customer.

Error Handling / Response to Abnormal Situations

If any of the above validations fail, appropriate error messages will be displayed. Orders cannot be processed for a customer whose data is not present in the system.

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3.3 Performance Requirements

- The system must support 4 to 5 simultaneous users without degradation in performance. Support for a higher number of concurrent users is planned for a future release.
- Invoice PDF generation must complete within 5 seconds of a successful order being processed under normal operating conditions.
- Stock level updates must be reflected in the system immediately upon successful order processing — no manual refresh required.
- Low stock alerts must be generated within 10 seconds of a stock level falling below the reorder threshold.
- Power BI dashboards should refresh data within a reasonable time as per Power BI's standard refresh capabilities.

3.4 Design Constraints

- The software is intended to be installed and operated on Microsoft Windows 10 and above only.
- The RDBMS to be used is PostgreSQL 14 or above only.
- The system must comply with applicable data protection practices, ensuring that customer details, product data, and financial records are securely handled and not exposed to unauthorised users.
- Wherever possible, development shall utilise open-source technologies (Node.js, PostgreSQL, Python, React.js, Electron.js) to reduce licensing costs and improve flexibility.
- The application must be designed with scalability and maintainability in mind, allowing it to accommodate increased user load and future feature additions without major architectural changes.
- The Administrator will be responsible for implementing a security policy to safeguard inventory and billing data from unauthorised access or modification.

3.5 Software System Attributes

3.5.1 Security

The application will be password-protected. Users will be required to enter a correct username and password to access the application. Each user will have access only to the modules corresponding to their assigned role. Users are isolated from one another's data and functions. After three consecutive failed login attempts, the account will be locked and the Admin notified. The Admin is responsible for implementing a broader security policy to safeguard inventory and billing data from unauthorised access or modification.

3.5.2 Maintainability

The application will be designed in a maintainable and modular manner. It will be straightforward to incorporate new requirements into individual modules as well as to add new modules in future releases. The use of open-source technologies (Node.js, PostgreSQL, Python, React.js) ensures long-term community support and ease of maintenance.

3.5.3 Portability

The application is designed for Windows 10 and above. The PostgreSQL database backend is cross-platform and can be migrated to a Linux-based server in future releases if required without changes to the core application logic.

3.5.4 Reliability

The Mean Time Between Failures (MTBF) of the software should not be less than 6 months. The system must handle erroneous input gracefully without crashing, displaying appropriate error messages to the user at all times.

3.5.5 Availability

The system must be available during standard retail store operating hours. Any scheduled maintenance or downtime should be communicated to users in advance by the system administrator.

3.6 Logical Database Requirements

The following tables will be maintained in the PostgreSQL database. Table definitions and schema are to be provided by the development team to the database administrator for setup.

Data Dictionary-

- User:

Field Name	Data Type	Key	Constraint	Description
user_id (PK)	INT	PK	NOT NULL, AUTO_INCREMENT, UNIQUE	Unique identifier for each user
username	VARCHAR(50)		NOT NULL, UNIQUE	Login username for the user
password_hash	VARCHAR(255)		NOT NULL	Bcrypt-hashed password string
role	ENUM		NOT NULL	Admin / Warehouse_Manager / General_Manager / Accountant
account_status	BOOLEAN		NOT NULL, DEFAULT TRUE	TRUE = active, FALSE = deactivated

- Customer:

Field Name	Data Type	Key	Constraint	Description
customer_id (PK)	INT	PK	NOT NULL, AUTO_INCREMENT, UNIQUE	Unique identifier for each customer
customer_name	VARCHAR(120)		NOT NULL	Full name of the customer
contact_number	VARCHAR(20)		NULL	Primary contact phone number
address	TEXT		NULL	Delivery or billing address
email	VARCHAR(150)		NULL, UNIQUE	Customer email address

- Product:

Field Name	Data Type	Key	Constraint	Description
product_id (PK)	INT	PK	NOT NULL, AUTO_INCREMENT, UNIQUE	Unique identifier for each product
sku	VARCHAR(50)		NOT NULL, UNIQUE	Stock Keeping Unit code
product_name	VARCHAR(150)		NOT NULL	Full descriptive name of the product
unit_price	DECIMAL(10,2)		NOT NULL, >= 0	Selling price per unit (excluding GST)
reorder_threshold	INT		NOT NULL, DEFAULT 10	Stock level at which a low-stock alert is triggered
current_stock_qty	INT		NOT NULL, >= 0	Current quantity available in warehouse

- Order:

Field Name	Data Type	Key	Constraint	Description
order_id (PK)	INT	PK	NOT NULL, AUTO_INCREMENT, UNIQUE	Unique identifier for each order
customer_id (FK)	INT	FK	NOT NULL, REFERENCES CUSTOMER	Customer who placed this order
order_datetime	DATETIME		NOT NULL, DEFAULT NOW()	Date and time the order was created
order_status	ENUM		NOT NULL, DEFAULT 'Pending'	Pending / Processing / Shipped / Delivered / Cancelled

- Invoice:

Field Name	Data Type	Key	Constraint	Description
invoice_id (PK)	INT	PK	NOT NULL, AUTO_INCREMENT, UNIQUE	Unique identifier for each invoice
order_id (FK)	INT	FK	NOT NULL, UNIQUE, REFERENCES ORDER	One invoice per order (1:1 relationship)
invoice_number	VARCHAR(30)		NOT NULL, UNIQUE	Human-readable invoice reference number
total_amount	DECIMAL(12,2)		NOT NULL, >= 0	Total payable amount including tax
pdf_path	VARCHAR(255)		NULL	File path or URL to the generated invoice PDF

- Order Items:

Field Name	Data Type	Key	Constraint	Description
order_item_id (PK)	INT	PK	NOT NULL, AUTO_INCREMENT, UNIQUE	Unique identifier for each order line item
order_id (FK)	INT	FK	NOT NULL, REFERENCES ORDER	Order this line item belongs to
product_id (FK)	INT	FK	NOT NULL, REFERENCES PRODUCT	Product being ordered
quantity	INT		NOT NULL, > 0	Number of units ordered
unit_price_at_time	DECIMAL(10,2)		NOT NULL, >= 0	Product price at the time of order (snapshot)
gst_rate_at_time	DECIMAL(5,2)		NOT NULL, >= 0	GST rate applied at time of order (snapshot)

- Low Stock Alert:

Field Name	Data Type	Key	Constraint	Description
alert_id (PK)	INT	PK	NOT NULL, AUTO_INCREMENT, UNIQUE	Unique identifier for each alert record
product_id (FK)	INT	FK	NOT NULL, REFERENCES PRODUCT	Product that triggered the alert
stock_at_trigger	INT		NOT NULL, >= 0	Stock quantity recorded when alert was triggered
reorder_threshold	INT		NOT NULL	Threshold value at the time the alert was created

3.7 Other Requirements

None at this stage.

Use Case Descriptions

This section provides simplified use case descriptions for each major system function. Each use case follows a consistent structure comprising a brief description, actors, pre-conditions, basic flow of events, alternative flows, and post-conditions.

UC-1. Login

UC-1.1 Brief Description

This use case describes how an authorised user logs into the Inventory Management System.

UC-1.2 Actors

- Admin
- General Manager
- Warehouse Manager
- Accountant

UC-1.3 Pre-Conditions

- The user must have a valid user account (username, password, and role) created in the system by the Admin or General Manager prior to login.
- The IMS application must be running on the client machine.

UC-1.4 Basic Flow

1. The system displays the Login Screen requesting the user to enter their username and password.
2. The user enters their username and password and submits the form.
3. The system validates the entered credentials against the user account database.
4. Upon successful validation, the system retrieves the user's role and displays the Role-Based Function Menu relevant to that role.
5. The user is now logged into the system and can proceed to use the available modules.

UC-1.5 Alternative Flows

UC-1-A1 Invalid Credentials: If the username or password entered is incorrect, the system displays an appropriate error message without revealing which field is wrong. The user may re-enter their credentials or cancel the login, at which point the use case ends.

UC-1-A2 Account Locked: If the user fails to log in three consecutive times, the system locks the account and displays a message notifying the user to contact the Admin. The use case ends.

UC-1.6 Post-Conditions

- If the use case was successful, the user is logged into the system and has access to modules corresponding to their role.
- If the use case was unsuccessful, the system state remains unchanged.

UC-1.7 Special Requirements

None.

UC-1.8 Extension Points

None.

UC-2. Manage User Accounts

UC-2.1 Brief Description

This use case allows the Admin or General Manager to create, modify, and delete user accounts within the system.

UC-2.2 Actors

- Admin
- General Manager

UC-2.3 Pre-Conditions

- The Admin or General Manager must be logged into the system.
- At system startup, only the default Admin account exists.

UC-2.4 Basic Flow

1. The user navigates to the User Management Screen from the Role-Based Function Menu.
2. The system displays the list of existing user accounts and prompts the user to select an action: Add User, Update User, or Delete User.
3. Add User: The system prompts the user to enter Full Name, Username, Password, and Role. Upon submission, the new user account is created and a confirmation message is displayed.
4. Update User: The user selects an existing account. The system retrieves and displays the account details. The user makes the desired changes and submits. The system updates the record.
5. Delete User: The user selects an existing account. The system prompts for confirmation of deletion. Upon confirmation, the user account is deleted from the system.

UC-2.5 Alternative Flows

UC-2-A1 Duplicate Username: If the username entered during Add User already exists in the system, the system displays an error message and prompts the user to enter a different username.

UC-2-A2 User Not Found: If during Update or Delete, the selected user account does not exist, the system displays an error message and returns to the User Management Screen.

UC-2-A3 Action Cancelled: If the user cancels any action at any point, the system discards any unsaved changes and returns to the User Management Screen. The system state remains unchanged.

UC-2.6 Post-Conditions

- If the use case was successful, the user account is added, updated, or deleted as requested.
- If the use case was unsuccessful or cancelled, the system state remains unchanged.

UC-2.7 Special Requirements

None.

UC-2.8 Extension Points

None.

UC-3. Manage Products / Inventory

UC-3.1 Brief Description

This use case allows the Warehouse Manager to add, update, and delete product records in the inventory and monitor current stock levels.

UC-3.2 Actors

- Warehouse Manager

UC-3.3 Pre-Conditions

- The Warehouse Manager must be logged into the system.
- The Product Management module must be accessible to the Warehouse Manager role.

UC-3.4 Basic Flow

1. The user navigates to the Product Management Screen from the Role-Based Function Menu.
2. The system displays the current product/inventory list and prompts the user to select an action: Add Product, Update Product, or Delete Product.
3. Add Product: The system prompts the user to enter Product Code (SKU), Product Name, Product Category, Unit Price (MRP), GST Rate (%), Current Stock Quantity, and Reorder Threshold. Upon submission, the product is added to the system and a confirmation message is displayed.
4. Update Product: The user selects an existing product. The system retrieves and displays the product details. The user makes the desired changes and submits. The system updates the product record.
5. Delete Product: The user selects an existing product. The system prompts for confirmation of deletion. Upon confirmation, the product record is deleted from the system.

UC-3.5 Alternative Flows

UC-3-A1 Duplicate Product Code: If the Product Code (SKU) entered during Add Product already exists, the system displays an error message and prompts the user to enter a unique Product Code.

UC-3-A2 Product Not Found: If during Update or Delete, the selected product does not exist in the system, the system displays an error message and returns to the Product Management Screen.

UC-3-A3 Validation Failure: If any field fails validation (e.g., negative price, non-numeric quantity), the system displays an appropriate error message and does not save the record until all fields are valid.

UC-3-A4 Action Cancelled: If the user cancels any action, the system discards unsaved changes and returns to the Product Management Screen.

UC-3.6 Post-Conditions

- If the use case was successful, the product information is added, updated, or deleted from the system as requested.
- If the use case was unsuccessful or cancelled, the system state remains unchanged.

UC-3.7 Special Requirements

None.

UC-3.8 Extension Points

None.

UC-4. Process Order and Update Stock

UC-4.1 Brief Description

This use case describes how the system receives order data from the backend, validates the order, updates stock levels, and triggers invoice generation automatically.

UC-4.2 Actors

- System (automated backend process)
- Warehouse Manager (monitoring)

UC-4.3 Pre-Conditions

- Product records with current stock levels and GST rates must exist in the system.
- Customer data (name, contact, address) must be present in the system for the ordering customer.
- The order data (customer details and list of items with quantities) must be available at the backend.

UC-4.4 Basic Flow

1. The backend receives order data comprising customer identification, a list of Product Codes (SKUs), and their respective quantities.
2. The system validates each Product Code against the product database to confirm it exists.
3. The system checks that the requested quantity for each product does not exceed the available stock.
4. Upon successful validation, the system decrements the stock level for each ordered product by the respective ordered quantity.
5. The system records the order in the database with order details, date, and time.
6. The system triggers the invoice generation process (see UC-5).
7. If any product's updated stock level falls below its Reorder Threshold, the system triggers a low stock alert for the Warehouse Manager (see UC-6).

UC-4.5 Alternative Flows

UC-4-A1 Invalid Product Code: If a Product Code in the order does not exist in the system, the system logs the error, notifies the relevant user, and does not process the order.

UC-4-A2 Insufficient Stock: If the requested quantity for any product exceeds available stock, the system notifies the relevant user and does not process the order until the issue is resolved.

UC-4.6 Post-Conditions

- If the use case was successful, stock levels are updated, the order is recorded in the database, and invoice generation is triggered.
- If validation fails, no stock changes are made and the system state remains unchanged.

UC-4.7 Special Requirements

None.

UC-4.8 Extension Points

None.

UC-5. Generate Invoice

UC-5.1 Brief Description

This use case describes how the system automatically generates a GST-compliant invoice in PDF format upon successful order processing, and allows authorised users to view and print it.

UC-5.2 Actors

- System (automated)
- Admin
- General Manager
- Warehouse Manager
- Accountant (view/print)

UC-5.3 Pre-Conditions

- A valid order must have been successfully processed by the system (UC-4 completed successfully).
- Product records with valid GST rates must be present.
- The Python PDF generation module must be operational.

UC-5.4 Basic Flow

1. Upon successful order processing, the system automatically invokes the Python PDF generation module.
2. The system compiles all order data: Invoice Number (system-generated), Invoice Date and Time, Customer Name, Contact, and Address, list of ordered products with quantities and unit prices, GST amounts per product, and Total Amount Payable.
3. The Python module generates a GST-compliant PDF invoice and saves it to the designated output directory.
4. The system stores a reference to the generated invoice in the database linked to the corresponding order.
5. Authorised users (Admin, General Manager, Warehouse Manager, Accountant) can navigate to the Invoice Management Screen to view and/or print the invoice.

UC-5.5 Alternative Flows

UC-5-A1 PDF Generation Failure: If the Python PDF module fails during generation, the system logs the error and notifies the user. The order record is preserved in the database to allow for manual re-generation of the invoice.

UC-5-A2 Invoice Not Found: If a user attempts to view an invoice that cannot be retrieved from the database, the system displays an appropriate error message.

UC-5.6 Post-Conditions

- If the use case was successful, a GST-compliant PDF invoice is generated, saved, and available for viewing and printing by authorised users.
- If the use case failed, the order data remains intact in the database for recovery.

UC-5.7 Special Requirements

None.

UC-5.8 Extension Points

None.

UC-6. Low Stock Alert

UC-6.1 Brief Description

This use case describes how the system automatically monitors stock levels and notifies the Warehouse Manager when any product's stock falls below the defined Reorder Threshold.

UC-6.2 Actors

- System (automated)
- Warehouse Manager

UC-6.3 Pre-Conditions

- Product records with defined Reorder Thresholds must exist in the system.
- The Warehouse Manager must be logged into the system to view alerts.

UC-6.4 Basic Flow

1. Upon completion of each order processing cycle, the system checks the updated stock level of each product involved in the order.
2. If any product's current stock quantity is found to be strictly less than its defined Reorder Threshold, the system generates a low stock alert for that product.
3. The alert is stored in the system and displayed on the Warehouse Manager's Low Stock Alerts Screen.
4. The Low Stock Alerts Screen displays the Product Code, Product Name, Current Stock Quantity, and Reorder Threshold for each affected product.
5. The Warehouse Manager reviews the alerts and takes appropriate action (e.g., initiating stock replenishment).

UC-6.5 Alternative Flows

UC-6-A1 No Alerts: If no product's stock level falls below the Reorder Threshold after order processing, no alert is generated and the Low Stock Alerts Screen displays a message confirming that all stock levels are within acceptable limits.

UC-6-A2 Duplicate Alert Prevention: If an active low stock alert already exists for a product, the system will not generate a duplicate alert until the stock level is replenished above the Reorder Threshold.

UC-6.6 Post-Conditions

- If the use case was successful, low stock alerts are displayed to the Warehouse Manager for all products requiring attention.
- If no products are below threshold, the system state remains unchanged and no alerts are shown.

UC-6.7 Special Requirements

None.

UC-6.8 Extension Points

None.

UC-7. View Power BI Analytics Dashboard

UC-7.1 Brief Description

This use case allows the Admin and General Manager to access the Power BI analytics dashboard embedded within the IMS, providing insights into sales trends, inventory levels, product demand, and customer demographics.

UC-7.2 Actors

- Admin
- General Manager

UC-7.3 Pre-Conditions

- The user must be logged in as Admin or General Manager.
- Power BI must be configured and connected to the PostgreSQL data source.
- Sufficient order, inventory, and customer data must exist in the database for meaningful analytics.

UC-7.4 Basic Flow

1. The user navigates to the Power BI Dashboard Screen from the Role-Based Function Menu.
2. The system loads the embedded Power BI dashboard, connecting to the PostgreSQL data source.
3. The dashboard displays visual reports including: sales trends (daily, weekly, monthly), product-wise demand and revenue, current inventory status and stock movement, customer demographic insights (age, location), and profit margins per product.
4. The user interacts with the dashboard filters and views to explore the data as required.
5. The user exits the dashboard when done.

UC-7.5 Alternative Flows

UC-7-A1 Power BI Connection Unavailable: If the Power BI service or the database connection is unavailable, the system displays an appropriate error message. Core IMS operations continue to function independently.

UC-7.6 Post-Conditions

- The user has viewed the relevant analytical reports. No data is modified by this use case.
- If the connection was unavailable, the system state remains unchanged.

UC-7.7 Special Requirements

None.

UC-7.8 Extension Points

None.

UC-8. Manage Customer Records

UC-8.1 Brief Description

This use case describes how customer data is captured, stored, and managed in the backend system for invoice generation and Power BI analytics purposes.

UC-8.2 Actors

- Admin
- General Manager
- System (data capture at order time)

UC-8.3 Pre-Conditions

- The Admin or General Manager must be logged into the system to view or manage customer records.
- Customer data is also populated automatically by the system at the time of order placement.

UC-8.4 Basic Flow

1. The user navigates to the Customer Records Screen from the Role-Based Function Menu.
2. The system displays the list of existing customer records.
3. The user may view individual customer records including Customer ID, Name, Contact Number, Address, Age, and Purchase History.
4. New customer records are captured automatically by the system when order data arrives containing customer details not already present in the database.
5. The Admin may update customer records where required, ensuring data accuracy for invoice generation and analytics.

UC-8.5 Alternative Flows

UC-8-A1 Customer Not Found: If an order arrives for a customer whose data is not present in the system, the system will capture and create a new customer record from the order data before proceeding with order processing.

UC-8-A2 Validation Failure: If customer data fails validation (e.g., non-numeric contact number, invalid age), the system displays an appropriate error message and does not save the record.

UC-8.6 Post-Conditions

- Customer records are accurately maintained in the database for use in invoice generation and Power BI analytics.
- If validation failed, no changes are made to the system.

UC-8.7 Special Requirements

None.

UC-8.8 Extension Points

None.